

12. (or)

12.1 $\Rightarrow$ Palmist use convex lens because it show enlarged & virtual & erect image.

12.2 $\Rightarrow$ betⁿ focus & centre of curvature of lens.

12.3 $\Rightarrow$ Here,

$$U = -5 \text{ cm}; f = 10 \text{ cm}$$

Using lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{10} = \frac{1}{v} - \frac{1}{-5}$$

$$\frac{1}{10} = \frac{1}{v} + \frac{1}{5}$$

$$\frac{1}{10} - \frac{1}{5} = \frac{1}{v}$$

$$\frac{1-2}{10} = \frac{1}{v}$$

$$\frac{-1}{10} = \frac{1}{v}$$

$$\therefore \boxed{v = -10 \text{ cm}}$$

$$m = \frac{v}{u}$$

$$= \frac{-10}{-5}$$

$$\boxed{m = 2}$$

(ii) a) Maxwell's Right hand Thumb Rule.

b) perpendicular

11.

11.1 $\Rightarrow$ Here, $V = 3V$; $R_1 = 10\Omega$; $R_2 = 15\Omega$

$$\text{Now, (i) } I_1 = \frac{V}{R_1} \\ = \frac{3}{10}$$

$$I_1 = 0.3 \text{ A}$$

$$\text{(ii) } I_2 = \frac{V}{R_2} \\ = \frac{3}{15}$$

$$I_2 = 0.2 \text{ A}$$

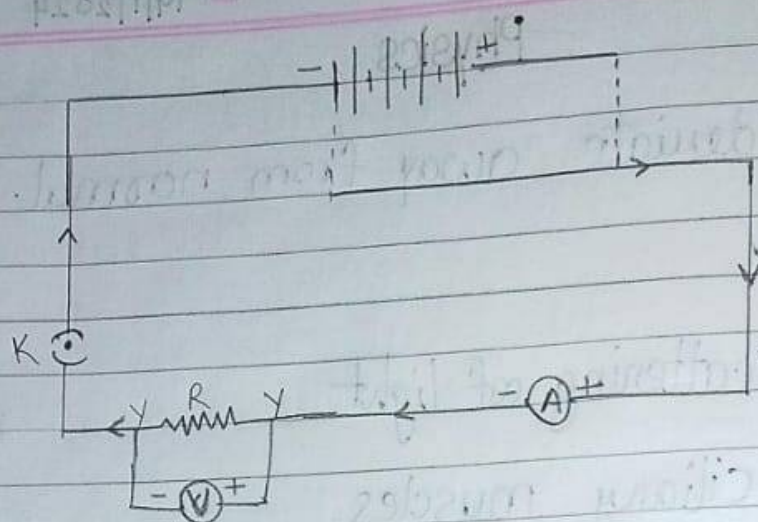
11.2 $\Rightarrow$ Straight Line (Uniform).

11.3 $\Rightarrow$ Slope of V-I Graph give resistance.

~~at the point~~

$\therefore$ Series combination has maximum resistance than other resistor.

Hence, The line C has max. slope which give series combination.



8. Here, $u = -25\text{ cm}$; $v = -75\text{ cm}$

Using lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\therefore \frac{1}{f} = \frac{1}{-75} - \frac{1}{-25}$$

$$\frac{1}{f} = -\frac{1}{75} + \frac{1}{25}$$

$$\frac{1}{f} = \frac{-1 + 3}{75}$$

$$\frac{1}{f} = \frac{2}{75}$$

$$f = \frac{75}{2} \text{ cm} \Rightarrow \underline{\underline{37.5 \text{ cm}}}$$

Now,

$$p = \frac{100}{f(\text{in cm})}$$

$$= \frac{100}{\frac{75}{2}} \Rightarrow \frac{200}{75}$$

$$[p \Rightarrow 2.67 \text{ D}]$$

$\therefore$ It is converging lens. (Convex).

9. (i) Resistance change.

$$\rightarrow L = 2L; A = \frac{A}{2}$$

$$\therefore R = \rho \frac{L}{A} \quad \text{--- (i)}$$

$$\therefore R' = \rho \frac{2L}{\frac{A}{2}}$$

$$\therefore R' = \rho \frac{4L}{A}$$

$$\therefore R' = 4 \left(\rho \frac{L}{A} \right)$$

$$\therefore [R' = 4R] \quad \left[\rho \frac{L}{A} = R \right]$$

$\therefore$ Resistance becomes 4 times.

(ii) Resistivity does not depend on length & Area. So, it does not change.

10. (i) By the use of right hand thumb rule, the direction of M.F. would anticlockwise around direction of current. So, M.F. at pt. P into the plane of paper & At A out from plane of paper. The magnetic field stronger near pt. A than pt. P.

Physics

1. (b) deviate away from normal.
2. (d) IV
3. (c) Scattering of light
4. (c) ciliary muscles.
5. (d) Assertion is false but Reason is true.
6. The ratio of velocity of light in vacuum to the velocity of light in the material or Refractive Index.

$$n = \frac{c}{v}$$

Here,

$$n = 1.5, v = 2 \times 10^8 \text{ ms}^{-1}$$

$$\therefore c = nv$$

$$= 2 \times 10^8 \times 1.5$$

$$\therefore \boxed{c = 3 \times 10^8}$$

7. The error of diagram was that ammeter connected in parallel & Voltmeter connected in series.